

# ECON 311

## Lecture 19: Covid Recession and International Transmission of Monetary Policy

## COVID

International transmission of monetary policy

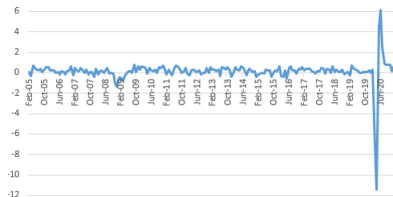
Conclusion

# Economic Indicators during Covid: RGDP

Canadian RGDP (Chained 2012 Dollars)  
1/2005-10/2021



% Change in RGDP



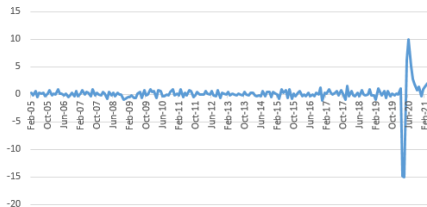
# RGDP– Cycle

Canadian RGDP Cycle 01/2005-10/2020

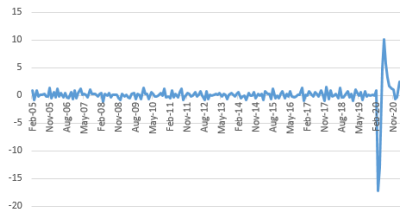


# Employment in Canada (Total Hours Worked)

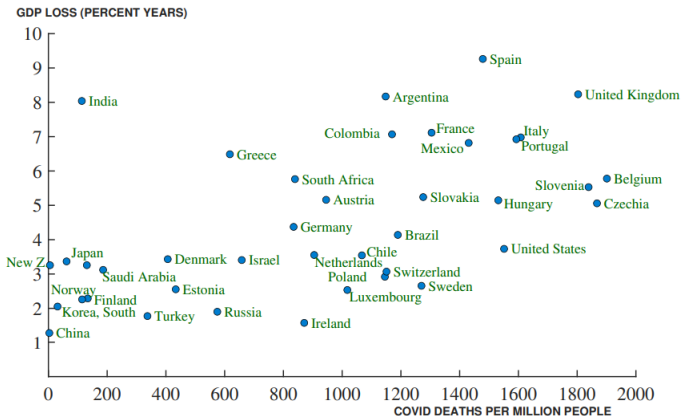
% Change in Total Hours Worked (all sectors)



% Change in Total Hours Worked (service)



# International Covid Shock



Note: "GDP Loss" is the *cumulative* loss in GDP since the start of 2020 and is annualized. For example, a value of 6 means that the loss since the start of 2020 is as if the economy lost six percent of its annual GDP. GDP data are through 2020Q3 and COVID-19 deaths are as of February 26, 2021.

# COVID Recession

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- Through the lens of our model, the COVID Recession is hard to analyse because:
  - ▶ Multiple Shocks to the economy (supply-side and demand-side)
  - ▶ (Short-run???) shocks to production capacity
    - ▶ Our framework allows for either short-run supply shocks to the firm's price setting behaviour (cost-push shocks) or long-run shocks to the production capacity.
    - ▶ Consensus is that it affected  $\bar{Y}$ , which affected demand parameters in the long-run with the short-run adjusting right away (did not see any deflation pressure — unlike overshooting in the IS-MP model)
    - ▶ As we recover, agents are failing to see that this is a more permanent effect on  $\bar{Y}$ , which is why we are currently seeing short-run overshooting.
    - ▶ In addition, we are experiencing a cost-push shock due to the war in Ukraine.

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# Exchange Rate

- The Real exchange can be expressed as follow:

$$RER = E_{C\$/U\$} \frac{P_{U\$}}{P_{C\$}}$$

- Assuming no transportation costs the exchange rate in the long-run is given by the law of one price:

$$E_{C\$/U\$} = \frac{P_{C\$}}{P_{U\$}}$$

- In the short run, exchange rates are set by demand for currency, for instance:

$$\uparrow i^{US} \Rightarrow \uparrow E \quad (\text{i.e. Canadian Dollar depreciates})$$

- With sticky prices:  $\uparrow E \Rightarrow \uparrow RER$

## Transmission of Monetary policy

- An change in nominal rates in the US has consequence for the Canadian Economy:

$$\uparrow i^{US} \Rightarrow \uparrow E \Rightarrow \uparrow RER \Rightarrow \uparrow EX, \downarrow IM$$

- This has an effect on the expenditure which results in an increase in output.
  - ▶ This effect is **similar** to a positive demand shock in our IS-MP and AS-AD model, we will formally study this effect in our SR models next class.

## Transmission of Monetary policy

- Impact of a change in nominal rates for the Canadian Economy:

$$\begin{aligned}\uparrow i^{US} &\Rightarrow \uparrow R^{US} \Rightarrow \uparrow E_{C\$/US\$} \Rightarrow \uparrow RER \Rightarrow \uparrow EX, \downarrow IM \\ \uparrow i^{CA} &\Rightarrow \uparrow R^{CA} \Rightarrow \downarrow E_{C\$/US\$} \Rightarrow \downarrow RER \Rightarrow \downarrow EX, \uparrow IM\end{aligned}$$

- To capture this effect, we write the net exports as:

$$NX = a_{NX} - b_{NX}(R_t - R_t^{US})$$

- This results in the following IS curve:

$$\tilde{Y}_t = \frac{1}{(1 - bc)} [\bar{a} - (\bar{b}_i + \bar{b}_{NX}) (R_t - \bar{r})] - 1$$

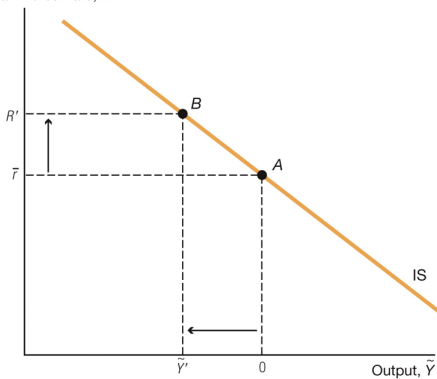
► where  $\bar{a} = \frac{a_c + a_i - b_i \bar{r} + G + a_{NX} - b_{NX} (\bar{r} - R_t^{US})}{\bar{Y}}$

# Transmission of Monetary policy

- The effect of domestic monetary policy is similar to before, but amplified. E.g.  $\uparrow R_t$ :

Increasing Interest Rates and the IS Curve

Real interest rate,  $R$

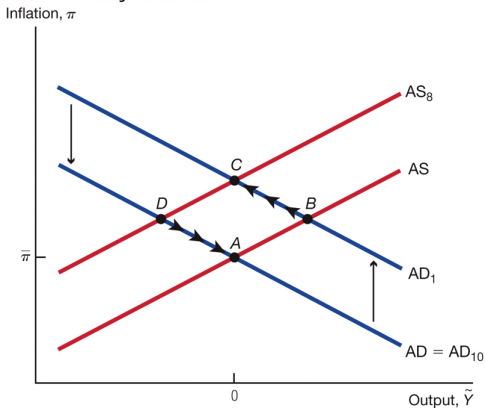


# Transmission of Monetary policy

- The effect of foreign monetary policy is like a demand shock. E.g. one time increase in  $R_t^{US}$ :

$$\uparrow R_t^{US} \Rightarrow \uparrow \bar{a}$$

An Increase in Foreign Interest Rates



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# Measuring Aggregate Economic Activity

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In this class, we study how economists measure aggregate economy activity.

- In particular, we studied
  1. different approaches to measure the value of output (expenditure, income and value added approach)
  2. how to compare economic activity across time (price indexes):
    - ▶ Chain-Weighting Approach to measure changes in real GDP over the long-run
  3. how to compare economic activity across countries:
    - ▶ using PPP-exchange rates

# Long-Run Economic Growth

- We also studied the determinants of long-run growth:
  1. Basic facts about long-run growth across countries
  2. Disaggregate cross-country differences in output per-capita into differences in inputs of an aggregate production function:
    - ▶  $Y = A K^{1/3} L^{2/3}$
  3. Study two important long-run growth models with very different implications for long-run growth:
    - ▶ Solow Growth Model
    - ▶ Romer Growth Model (Economics of Ideas)
- , and the long-run relation between money growth and inflation

# Short-Run Economic Fluctuations

- Lastly, we studied the determinants of short-run economic fluctuations:
  1. Basic facts about short-run deviations from potential output
  2. Shocks to the economy, inflation and government monetary policy (IS-MP model as well as standard and non-standard monetary policy instruments)
  3. The central macroeconomics model of short-run economic fluctuations (AS-AD model) with an application to the Great Recession, Covid Crisis and international transmission of monetary policy